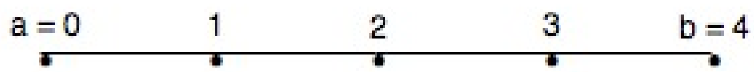
> $with(LinearAlgebra):$

Random Walk in One Dimension

First we compute the simple random walk with nodes 0,1,2,3,4 and absorbing state 4.



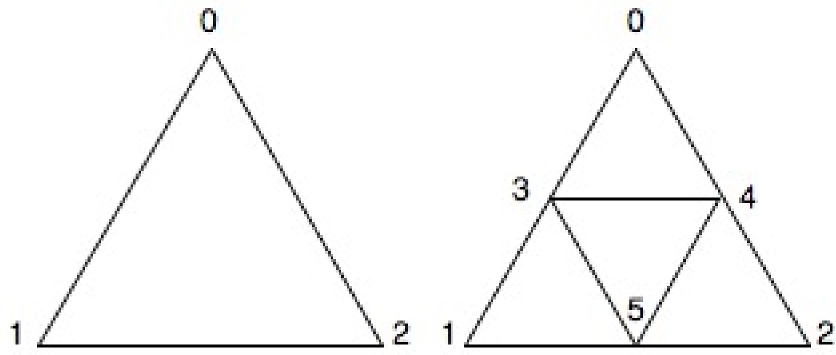
$$Q := \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{bmatrix} : \; T := \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - Q : \; N := T^{-1}$$

$$\begin{bmatrix} 4 & 6 & 4 & 2 \\ 3 & 6 & 4 & 2 \\ 2 & 4 & 4 & 2 \\ 1 & 2 & 2 & 2 \end{bmatrix} \tag{1}$$

$$t := MatrixVectorMultiply\left(N, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right)$$

$$\begin{bmatrix} 16 \\ 15 \\ 12 \\ 7 \end{bmatrix} \tag{2}$$

Random Walk on a Triangle



The vertices are 0 (top), 1 (bottom L), 2 (bottom R).;
The absorbing state is 0.;

$R1 := \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$: $Q1 := \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$: $\mathrm{T1} := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - Q1;$

$$\begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix} \tag{3}$$

$N1 := T1^{-1};$

$$\begin{bmatrix} \frac{4}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{4}{3} \end{bmatrix} \tag{4}$$

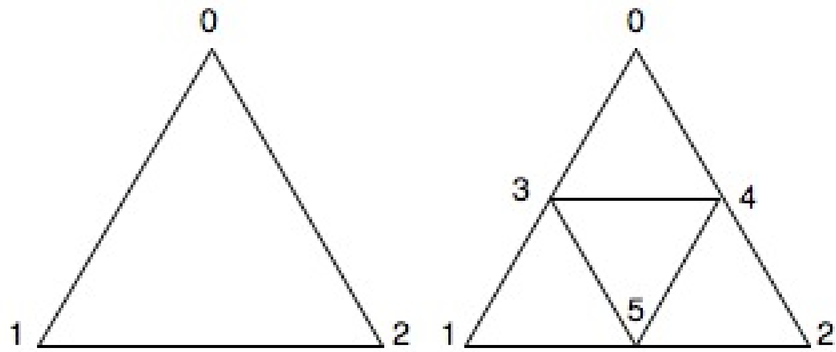
$t1 := MatrixVectorMultiply\left(N1, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} \tag{5}$$

$B1 := MatrixVectorMultiply(N1, R1)$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{6}$$

Random Walk on SG(2) prefractal approximation



Here we work with SG(2) and vertices 0 (top), 1 (bottom L), 2 (bottom R), 3 (middle L), 4 (middle R), 5 (middle Bottom),

Three absorbing states {0,1,2}.

$$R2a := \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{1}{4} \end{bmatrix} : \; Q2a := \begin{bmatrix} 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 \end{bmatrix} : \; T2a := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - Q2a;$$

$$\begin{bmatrix} 1 & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & 1 & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix} \quad (7)$$

$N2a := T2a^{-1};$

$$\begin{bmatrix} \frac{6}{5} & \frac{2}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{6}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{2}{5} & \frac{6}{5} \end{bmatrix} \quad (8)$$

$B2a := MatrixMatrixMultiply(N2a, R2a);$

$$\begin{bmatrix} \frac{2}{5} & \frac{2}{5} & \frac{1}{5} \\ \frac{2}{5} & \frac{1}{5} & \frac{2}{5} \\ \frac{1}{5} & \frac{2}{5} & \frac{2}{5} \end{bmatrix} \tag{9}$$

Two absorbing states {0,1}.

$R2c := \begin{bmatrix} 0 & 0 \\ \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{bmatrix} : \; Q2c := \begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 \end{bmatrix} : \; T2c := \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - Q2c;$

$$\begin{bmatrix} 1 & 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & 1 & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix} \tag{10}$$

$N2c := T2c^{-1};$

$$\begin{bmatrix} \frac{5}{3} & \frac{2}{3} & \frac{4}{3} & \frac{4}{3} \\ \frac{1}{3} & \frac{4}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{26}{15} & \frac{14}{15} \\ \frac{2}{3} & \frac{2}{3} & \frac{14}{15} & \frac{26}{15} \end{bmatrix} \tag{11}$$

$t2c := MatrixVectorMultiply\left(N2c, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right);$

$$\begin{bmatrix} 5 \\ 3 \\ 4 \\ 4 \end{bmatrix} \tag{12}$$

$B2c := MatrixMatrixMultiply(N2c, R2c)$

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \\ \frac{3}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{3}{5} \end{bmatrix} \tag{13}$$

One absorbing state {0}.

$$R2b := \begin{bmatrix} 0 \\ 0 \\ \frac{1}{4} \\ \frac{1}{4} \\ 0 \end{bmatrix} : \; Q2b := \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 \end{bmatrix} : \; T2b := \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} - Q2b;$$

$$\begin{bmatrix} 1 & 0 & -\frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{4} & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & -\frac{1}{4} & 1 & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix} \tag{14}$$

$N2b := T2b^{-1};$

$$\begin{bmatrix} \frac{20}{9} & \frac{10}{9} & \frac{20}{9} & \frac{16}{9} & \frac{8}{3} \\ \frac{10}{9} & \frac{20}{9} & \frac{16}{9} & \frac{20}{9} & \frac{8}{3} \\ \frac{10}{9} & \frac{8}{9} & \frac{22}{9} & \frac{14}{9} & 2 \\ \frac{8}{9} & \frac{10}{9} & \frac{14}{9} & \frac{22}{9} & 2 \\ \frac{4}{3} & \frac{4}{3} & 2 & 2 & \frac{10}{3} \end{bmatrix} \tag{15}$$

$$t2b := MatrixVectorMultiply\left(N2b, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}\right);$$

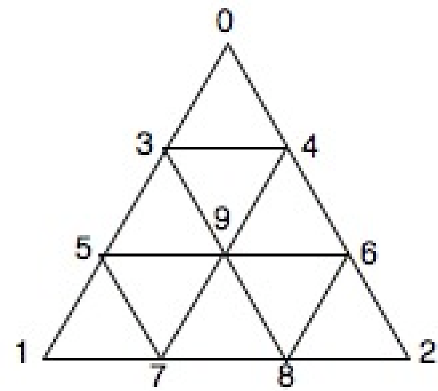
$$\begin{bmatrix} 10 \\ 10 \\ 8 \\ 8 \\ 10 \end{bmatrix} \tag{16}$$

$$B2b := MatrixVectorMultiply(N2b, R2b);$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \tag{17}$$

Random Walk on SG(3) prefractal approximation

Two absorbing states {0,1}.



$$R3b := \begin{bmatrix} 0 & 0 \\ \frac{1}{4} & 0 \\ \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \\ 0 & 0 \\ 0 & \frac{1}{4} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} : \quad Q3b := \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} \\ 0 & \frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ 0 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & 0 \end{bmatrix} :$$

$$T3b := \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} - Q3b;$$

$$\left[\begin{array}{cccccccc} 1 & 0 & 0 & 0 & -\frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & -\frac{1}{4} & 1 & 0 & -\frac{1}{4} & 0 & 0 & -\frac{1}{4} \\ 0 & -\frac{1}{4} & 0 & 1 & 0 & -\frac{1}{4} & 0 & -\frac{1}{4} \\ -\frac{1}{4} & 0 & -\frac{1}{4} & 0 & 1 & 0 & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 0 & -\frac{1}{4} & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & 0 & 0 & 0 & -\frac{1}{4} & -\frac{1}{4} & 1 & -\frac{1}{4} \\ 0 & -\frac{1}{6} & -\frac{1}{6} & -\frac{1}{6} & -\frac{1}{6} & -\frac{1}{6} & -\frac{1}{6} & 1 \end{array}\right] \tag{18}$$

$N3b := T3b^{-1}$

$$\left[\begin{array}{cccccccc} \frac{15}{7} & \frac{6}{7} & \frac{8}{7} & \frac{6}{7} & \frac{16}{7} & \frac{8}{7} & \frac{16}{7} & \frac{15}{7} \\ \frac{3}{7} & \frac{523}{315} & \frac{55}{63} & \frac{257}{315} & \frac{277}{315} & \frac{41}{63} & \frac{263}{315} & \frac{10}{7} \\ \frac{4}{7} & \frac{55}{63} & \frac{113}{63} & \frac{41}{63} & \frac{79}{63} & \frac{43}{63} & \frac{65}{63} & \frac{11}{7} \\ \frac{3}{7} & \frac{257}{315} & \frac{41}{63} & \frac{523}{315} & \frac{263}{315} & \frac{55}{63} & \frac{277}{315} & \frac{10}{7} \\ \frac{8}{7} & \frac{277}{315} & \frac{79}{63} & \frac{263}{315} & \frac{853}{315} & \frac{65}{63} & \frac{587}{315} & \frac{15}{7} \\ \frac{4}{7} & \frac{41}{63} & \frac{43}{63} & \frac{55}{63} & \frac{65}{63} & \frac{113}{63} & \frac{79}{63} & \frac{11}{7} \\ \frac{8}{7} & \frac{263}{315} & \frac{65}{63} & \frac{277}{315} & \frac{587}{315} & \frac{79}{63} & \frac{853}{315} & \frac{15}{7} \\ \frac{5}{7} & \frac{20}{21} & \frac{22}{21} & \frac{20}{21} & \frac{10}{7} & \frac{22}{21} & \frac{10}{7} & \frac{19}{7} \end{array}\right] \tag{19}$$

$$t3b := MatrixVectorMultiply\left(N3b, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}\right)$$

$$\begin{bmatrix} \frac{90}{7} \\ \frac{53}{7} \\ \frac{59}{7} \\ \frac{53}{7} \\ \frac{83}{7} \\ \frac{59}{7} \\ \frac{83}{7} \\ \frac{72}{7} \end{bmatrix} \qquad (20)$$

$$B3b := MatrixMatrixMultiply(N3b, R3b)$$

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{19}{30} & \frac{11}{30} \\ \frac{2}{3} & \frac{1}{3} \\ \frac{11}{30} & \frac{19}{30} \\ \frac{8}{15} & \frac{7}{15} \\ \frac{1}{3} & \frac{2}{3} \\ \frac{7}{15} & \frac{8}{15} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \tag{21}$$